

《Automotive Configuration》

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Lecture Schedule: Spring Semester Week 1~8

第二章 机体组及曲柄连杆机构

Chapter 2 Engine Block, Crank and Connecting Rod Mechanism



2.1 Overview

概述

2.2 Crank and Connecting Rod Mechanism

曲柄连杆机构工作原理

2.3 Engine Block

机体组

2.4 Cylinder Head

气缸盖

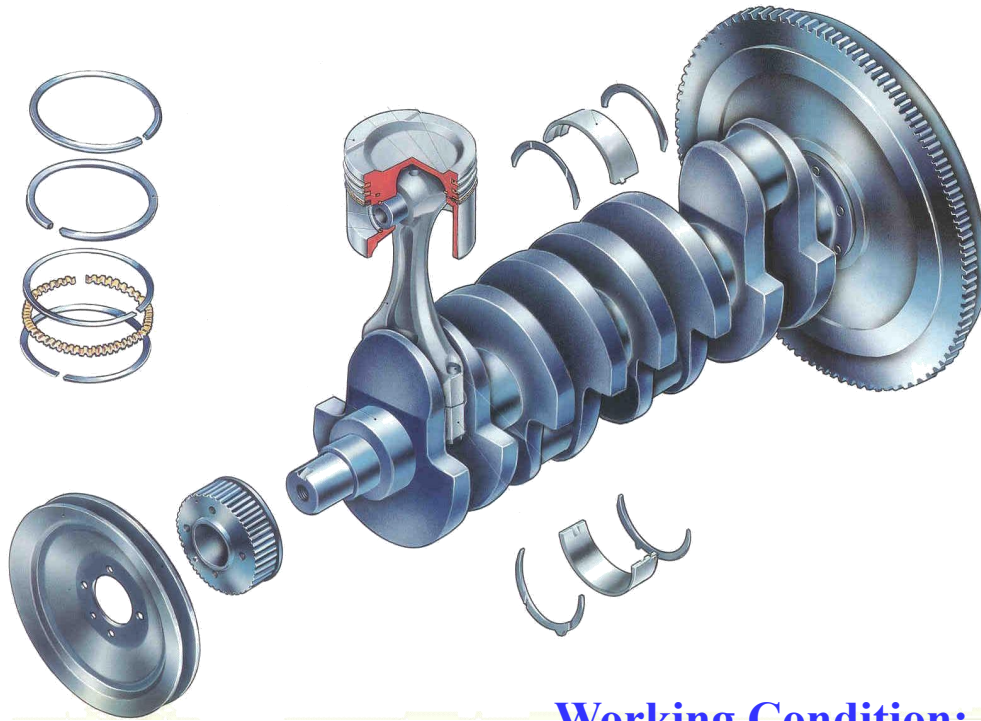
2.5 Piston-Rod Assemblies

活塞连杆组

2.6 Shaft-Flywheel Assemblies

曲轴飞轮组

2.1 Overview



Function:

The power system for reciprocating internal combustion engines uses a crank drive in which the connector rod end or the connecting rod joins the piston with the crankshaft (曲轴). The conrod converts the reciprocating (往复式) movement of the piston into rotary motion. Moreover, the conrod transfers forces from the piston to the crankshaft.

Component:

Engine block (机体组)

Piston-rod (活塞连杆组)

Shaft-flywheel (曲柄飞轮组)

Working Condition:

- ✓ Directly exposed to gas with high temperature and pressure inside the cylinder
- ✓ The inertia (惯性力) of the moving masses is relatively big due to high engine speed
- ✓ Corrosion (腐蚀) problem brought by high temperature exhaust gas
- ✓ Lack of lubrication (润滑).

2.2 Crank and Connecting Rod Mechanism (曲柄连杆组原理)



Working Condition:

High Temperature (up to 2500K)

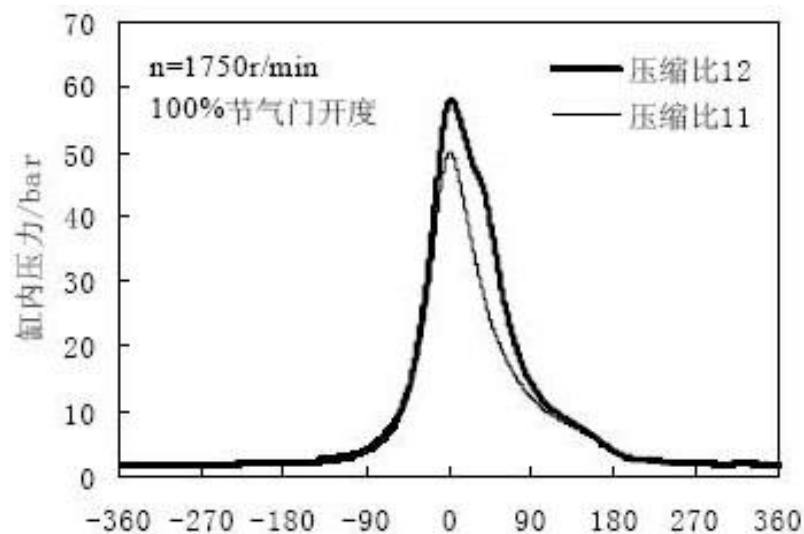
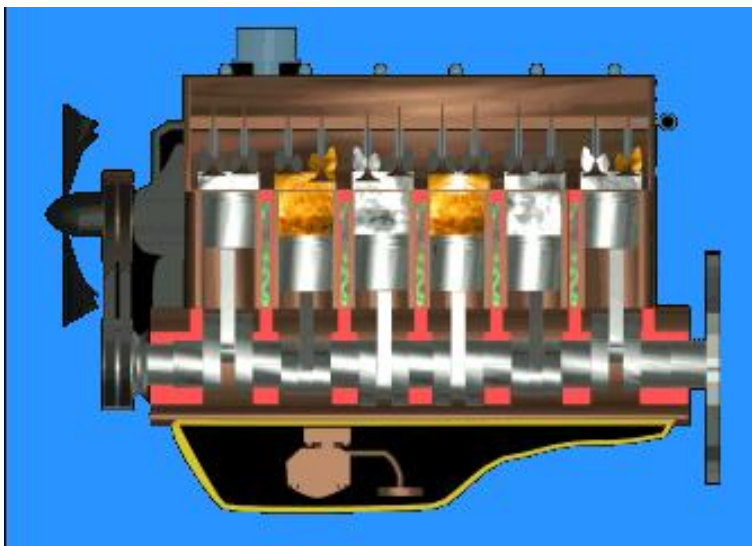
High Pressure (5~9MPa)

High Speed (3000~6000r/min) (100~200 stroke/s)

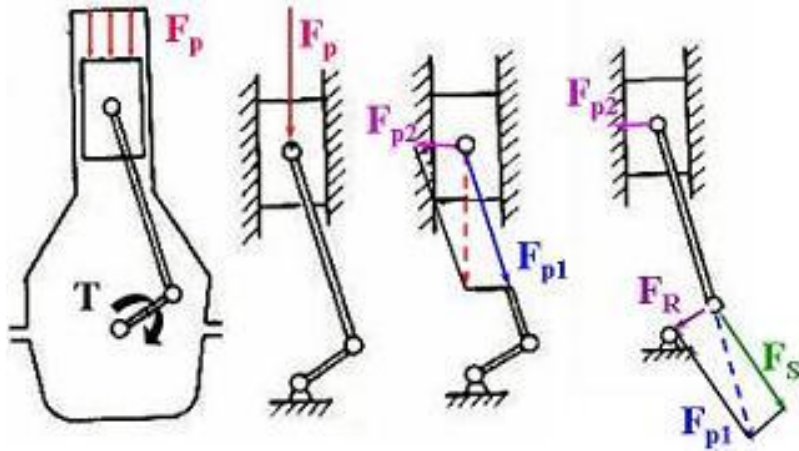
Chemical Corrosion (Cylinder, Cylinder head, Piston Components)

Loading:

Load under high pressure and variable motion, consist of gas force, inertia force, friction (摩擦), and other resistances



2.2.1 Gas forces



Working Stroke(做功冲程)

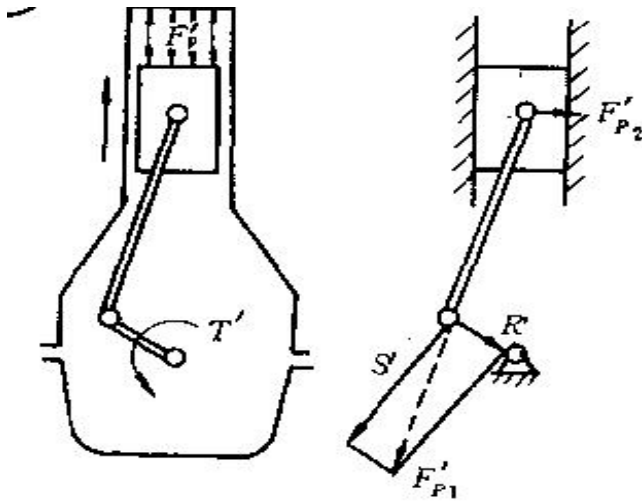
$F_p \rightarrow$ Piston $\rightarrow F_{p1}$ 、 F_{p2}

$F_{p1} \rightarrow$ conrod $\rightarrow F_R$ 、 F_S

$F_S \rightarrow$ Torque on crank $\rightarrow$ rotation

$F_{p2} \rightarrow$ Lateral Pressure (侧压力)

$\rightarrow$ Supported by engine block



Compression Stroke (压缩冲程)

$F'_p \rightarrow F'_{p1}$ 、 F'_{p2}

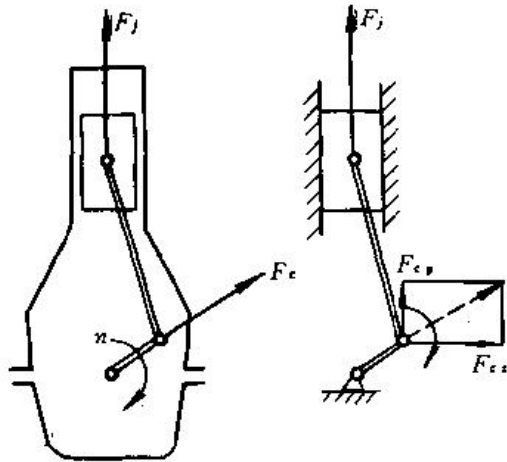
$F'_{p1} \rightarrow R'$ 、 S'

$R' \rightarrow$ Packing Force;

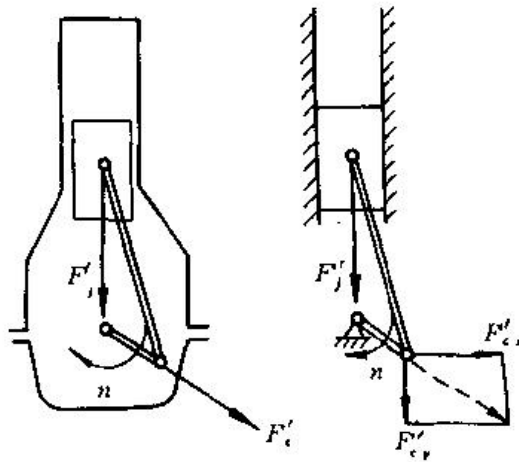
$S' \rightarrow$ anti-torque T' ,

$F'_{p2} \rightarrow$ Lateral Pressure

2.2.2 Inertia(惯性力) and centrifugal (离心力) Force



a) 活塞在上半行程时的惯性力



b) 活塞在下半行程时的惯性力

Inertia force:

- Generated during the linear reciprocating motion of piston(活塞), and Small-end eye(连杆小头)
- Changing law of piston velocity(from TDC to BDC):
0~accelerate~maximum~decelerate~0
- Piston moving downwards:

Accelerate the first half stroke, direction of inertia force points up

Decelerate the second half stroke, direction of inertia force points down

Centrifugal force:

- Generated during rotation of crank(曲柄), and big-end eye(连杆大头)
- Determined by crank radius, rotation mass, and engine speed

Larger mass of piston and connecting rod leads to bigger inertia force and centrifugal force, which would increase the vibration of engine and periodic load on components and bearing, which cause abrasion (磨损) and deformation (变形).

2.2.2 Inertia(惯性力) and centrifugal (离心力) Force

Conrod Masses for Various Applications

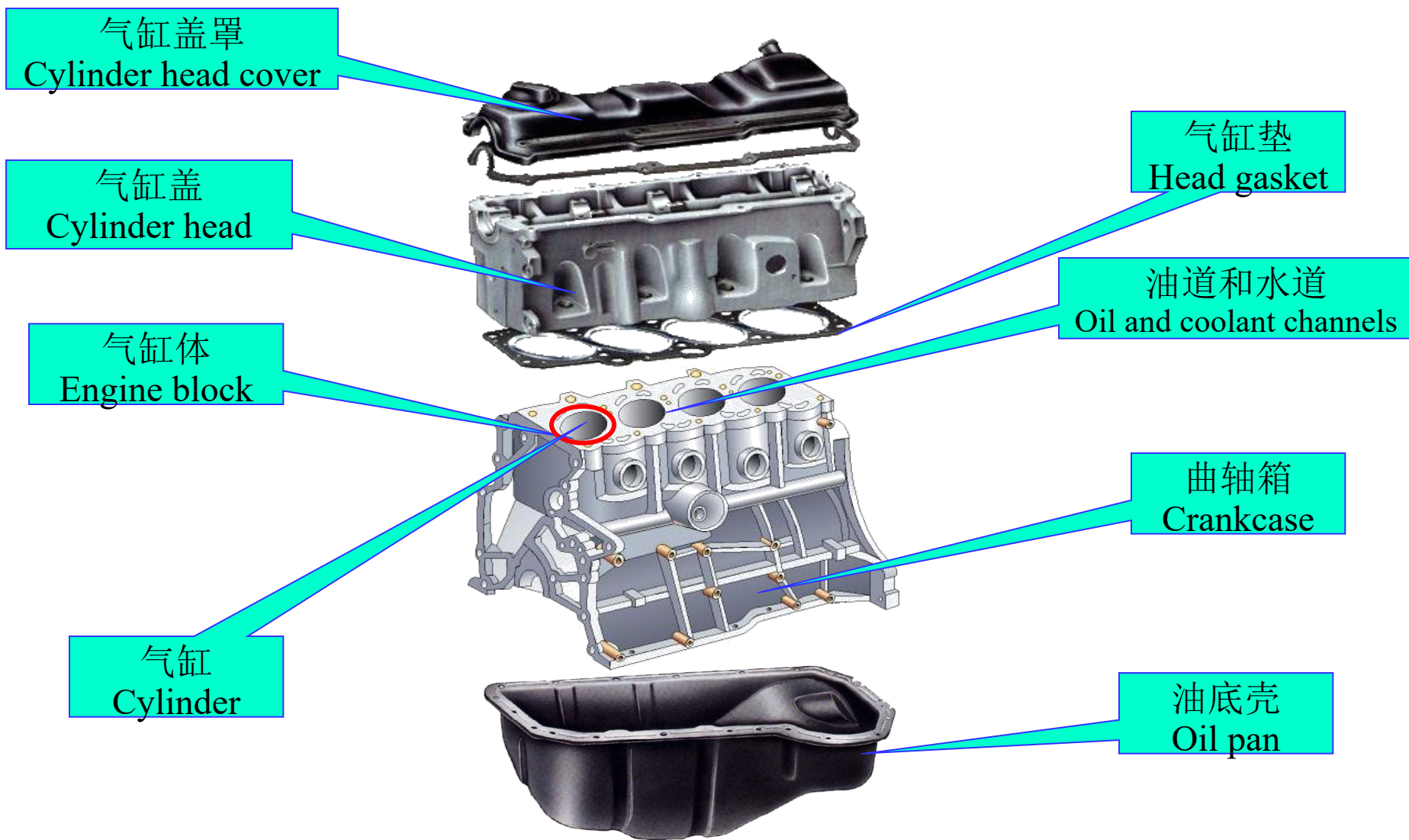
Application	Mass	Material
Mass production truck diesel	1.6 to 5 kg	Forged Steel(锻钢)
Mass production passenger car gasoline engine	0.4 to 1 kg	Forged Steel, gray casting (灰铸铁), sintered steel (烧结钢)
Sport use	0.4 to 0.7 kg	Steel, titanium
Racing engine/F1	0.3 to 0.4 kg	Titanium, carbon fiber
Compressor	0.2 to 0.6 kg	Aluminum

2.3 Engine Block (机体组)

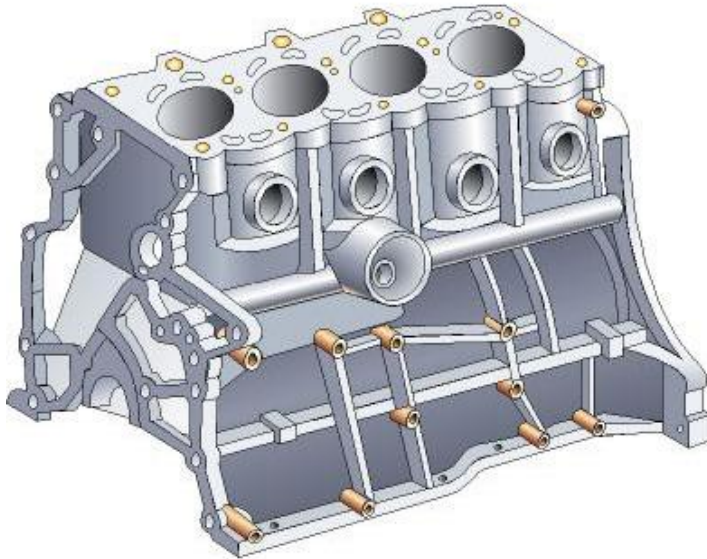
The primary functions that the engine block fulfills are:

- Absorbing the gas and mass forces in the crankshaft bearings (衬套) and at the cylinder head bolts.
- Accepting the energy conversion assembly, comprising the pistons, conrods, crankshaft, and flywheel.
- Carrying the crankshaft and (only rarely today) the camshaft(凸轮轴)
- Accepting channels to convey operating media, primarily lubricants and coolant.

Because of the variety of functions to be carried out, the engine block is subjected to differing types of loads that are superimposed one upon another. The engine block is the component that encloses the cylinders, the crankshaft, the cylinder head, and the head gasket (气缸垫).

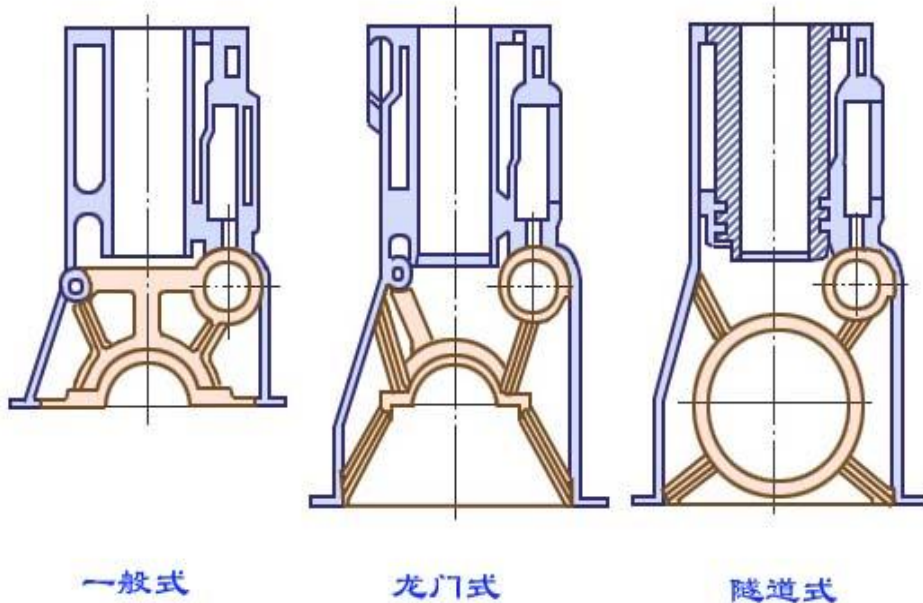


2.3.1 Cylinder Block (气缸体)



- Cylinder block consist of cylinder and upper half of crankshaft, and is usually made of gray casting.
- It transporting the coolant (冷却液) through cavities (腔) and channels contained in the engine block in liquid-cooled engines.
- Reinforcing rib (加强肋板) could be found on the cylinder block because the block is exposed to tensile (拉力) and compression loading, bending, and torsion (扭转) as a result of mass and gas forces.

2.3.1 Cylinder Block (气缸体)



The main bearing pedestal (轴承座) area in engine blocks is the area around the crankshaft bearings. The forces acting on the crankshaft bearings have to be taken up here. Cylinder blocks are divided into three types:

- Oil pan flange (油底壳) level with the center of the crankshaft (left)

Most widely used mass-production version with favorable manufacturing costs, however, less stiffness and less favorable acoustic properties

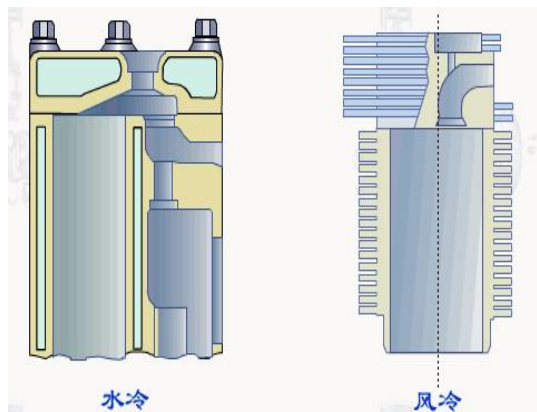
- Oil pan flange below the center of the crankshaft (middle)

Have the benefits of stiffness and acoustic properties, lack of manufacturability .

- The “tunnel” type (right)

Have the best stiffness of the three type, lack of manufacturability, hard at crank dismounting, not widely used

2.3.2 Cylinder Cooling and Arrangements (缸体冷却及排列形式)

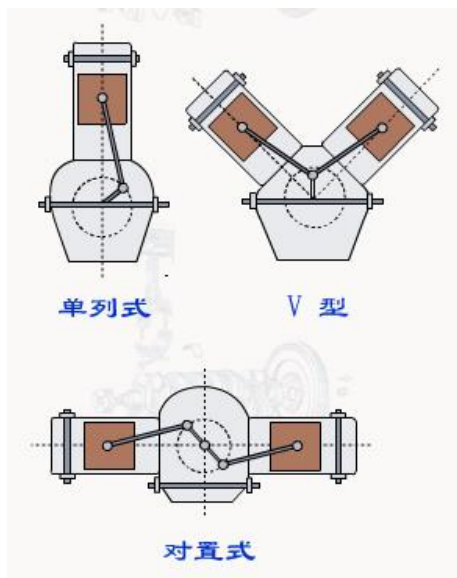


- Water cooling

With just a very exceptions today's automotive engines are water cooled. The cylinders are surrounded by a water-filled cavity, the water jacket or cooling jacket

- Air cooling

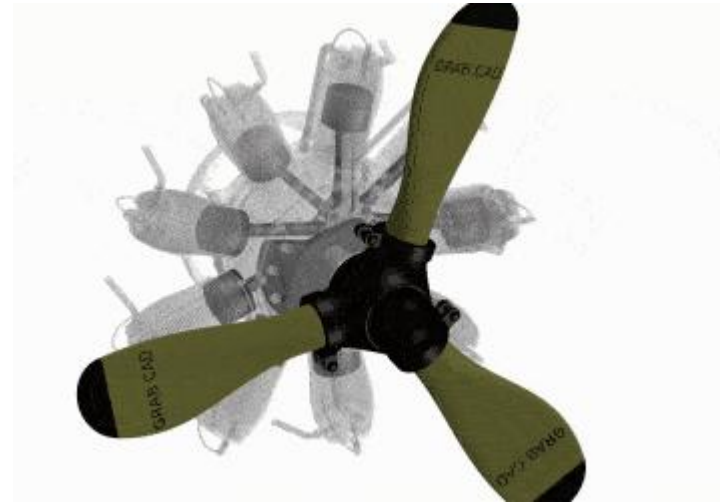
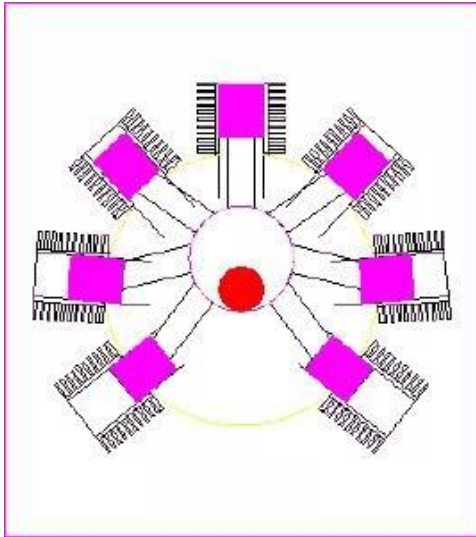
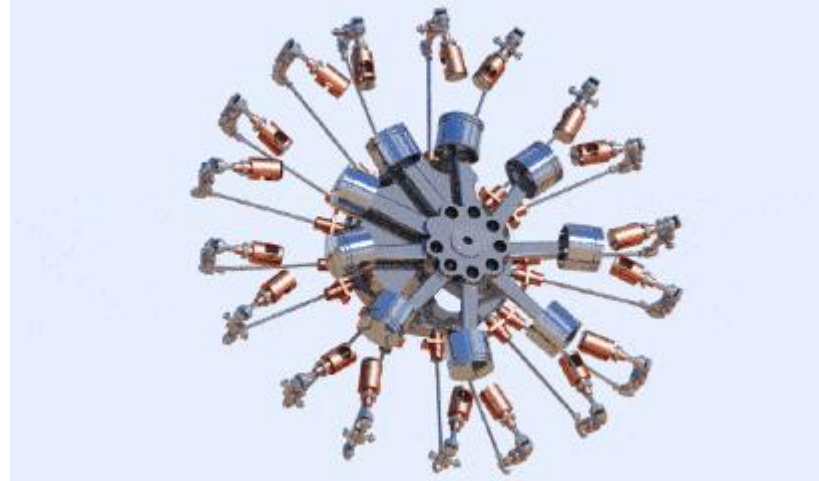
Heat dissipation in air-cooled cylinders is dependent upon the thermal conductivity of the cylinder fins ,shape of cooling fins, and the way in which cooling air passes across the fins, hardly used today



- Cylinder arrangement

Common cylinder arrangements are from 1 to 6 cylinders in-line(单列式) or from 2 to 16 cylinders in V-formation (V型). Flat engines (对置式) – like a V design flattened out – are common in small airplanes and motorcycles and were a hallmark of Volkswagen automobiles into the 1990s. Flat 6s are still used in many modern Porsches, as well as Subarus. Many flat engines are air-cooled. Less common, but notable in vehicles designed for high speeds is the W formation, similar to having 2 V engines side by side.

Radial Engine (星型发动机)



2.3.3 Cylinder Sleeves (汽缸套)

- **Wet Cylinder Sleeve (right)**

Wet cylinders are slid into the engine block, mating with mounting areas machined and prepared accordingly

- ✓ Advantage: Freedom in selecting material for sleeve, simple interchangeability and repairs

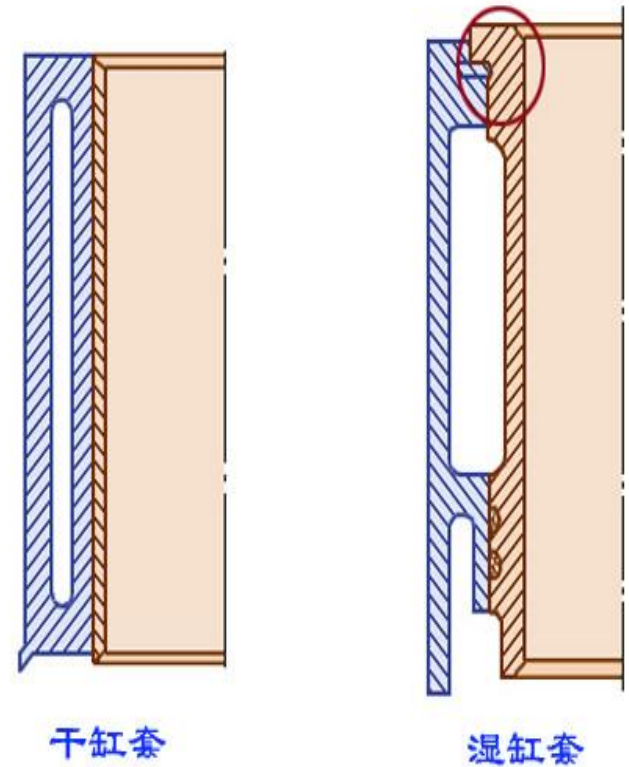
- ✓ Disadvantage: *higher manufacturing costs*

- **Dry Cylinder Sleeve (left)**

Dry sleeves are pressed, shrink fit, or cast in place in the engine block. The sleeves are inserted in the engine block mold, and the molten aluminum alloy is cast around them

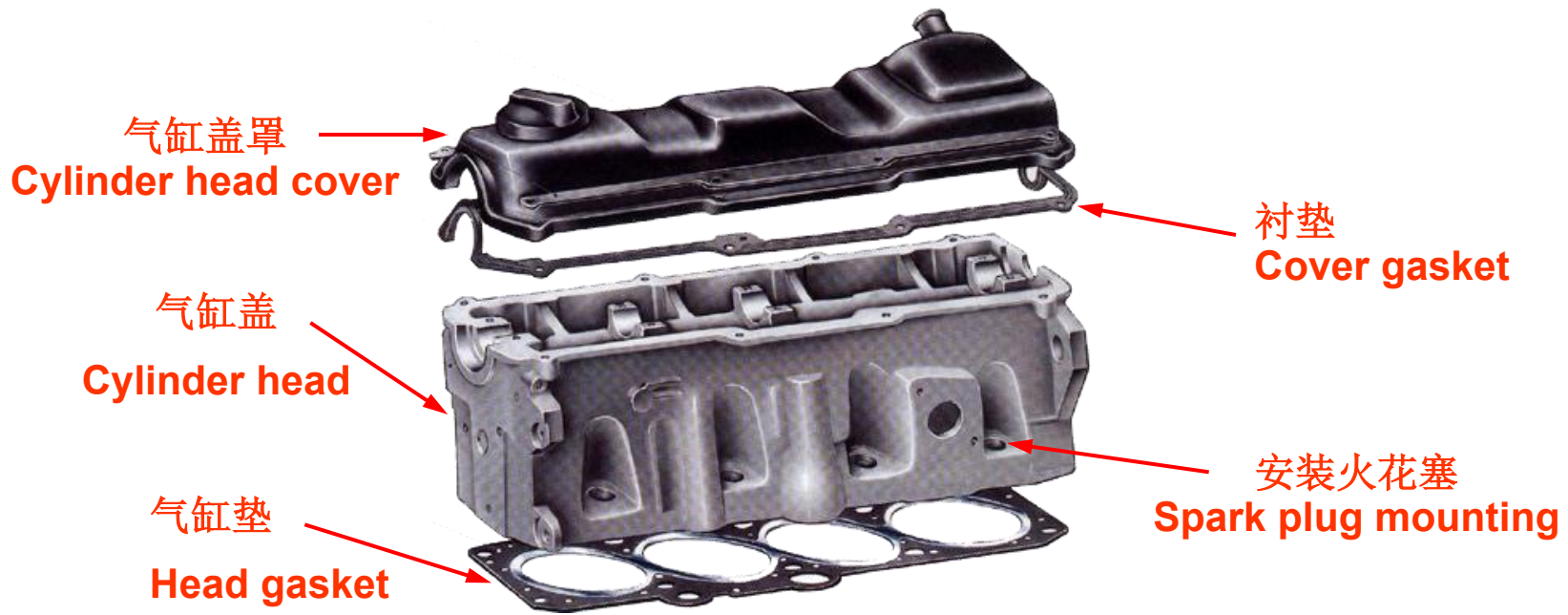
- ✓ Advantage: Easy repair option by reaming out to oversize dimensions, separate manufacture of cylinder finishing surface, and the option for combining sleeves with an engine block made of a different aluminum alloy, lower cost

- ✓ Disadvantage: *poorer heat transfer* between the cylinder running surface and the water jacket

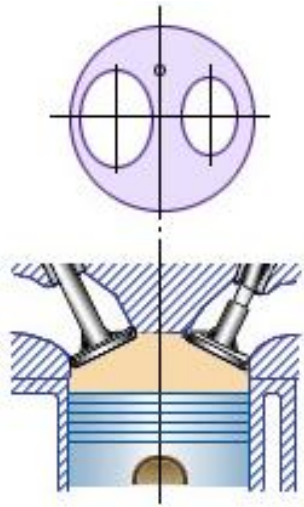


2.4 Cylinder Head (气缸盖)

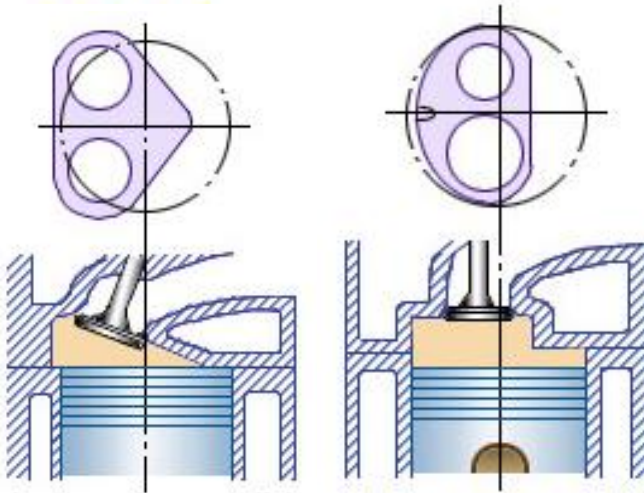
- Great importance is attached to cylinder head design and engineering during engine development. The cylinder head determines, like no other subassembly in the engine, operating properties such as performance level, torque, exhaust emissions, fuel consumption, and acoustic properties (声学性能)



2.4.1 Combustion Chamber (燃烧室)



半球形



楔形

盆形

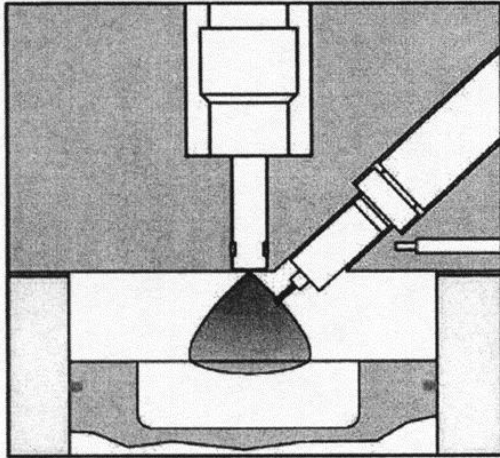
Cylinder head provide most part of the combustion chamber in SI engines (点燃式发动机), hence plays an important role in engine performance.

Combustion chamber for SI engines are usually divided into three types

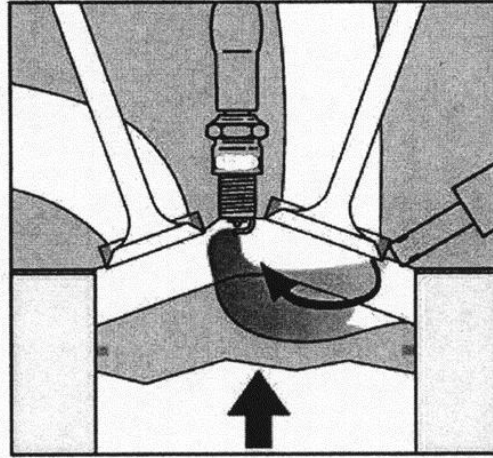
- Hemispheric: Compact structure with spark plug placed in the middle of combustion chamber. Combustion rate is higher, which leads to higher thermo efficiency. Widely used in automotive engines
- Pentroof: Simple structure with less heat loss, easy to generate tumble (滚流) during compression stroke, higher volumetric efficiency. However, spark plug is placed at top of combustion chamber, therefore, combustion rate is less higher than hemispheric type
- Basin shaped: Better manufacturability, lower cost

Combustion Chamber for GDI

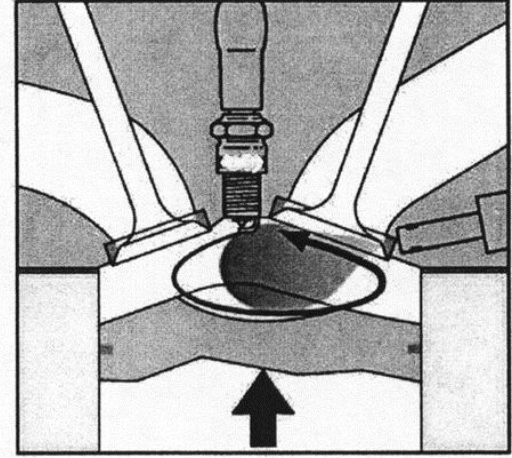
Spray-guided



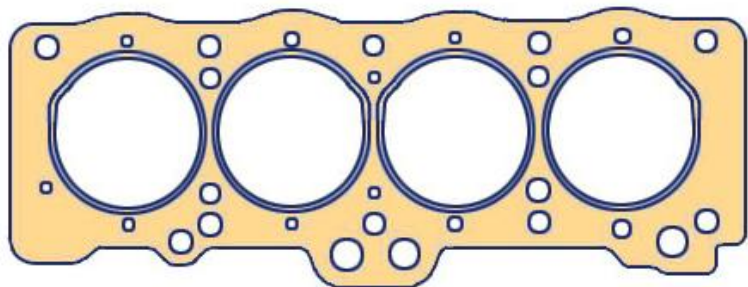
Wall-guided



Air-guided



2.4.2 Head Gasket and Oil Pan (汽缸垫及油底壳)



汽缸垫

The head gasket is becoming more important in modern engines.

- Function: Sealing off combustion chamber, cooling system, and oil passages, serves to transmit forces between the cylinder head and the engine block. Exerts considerable influence on force distribution within the entire assembly system
- Type: Ferrolastic(铁弹体), metal (金属), metaloflex layered (多层金属)



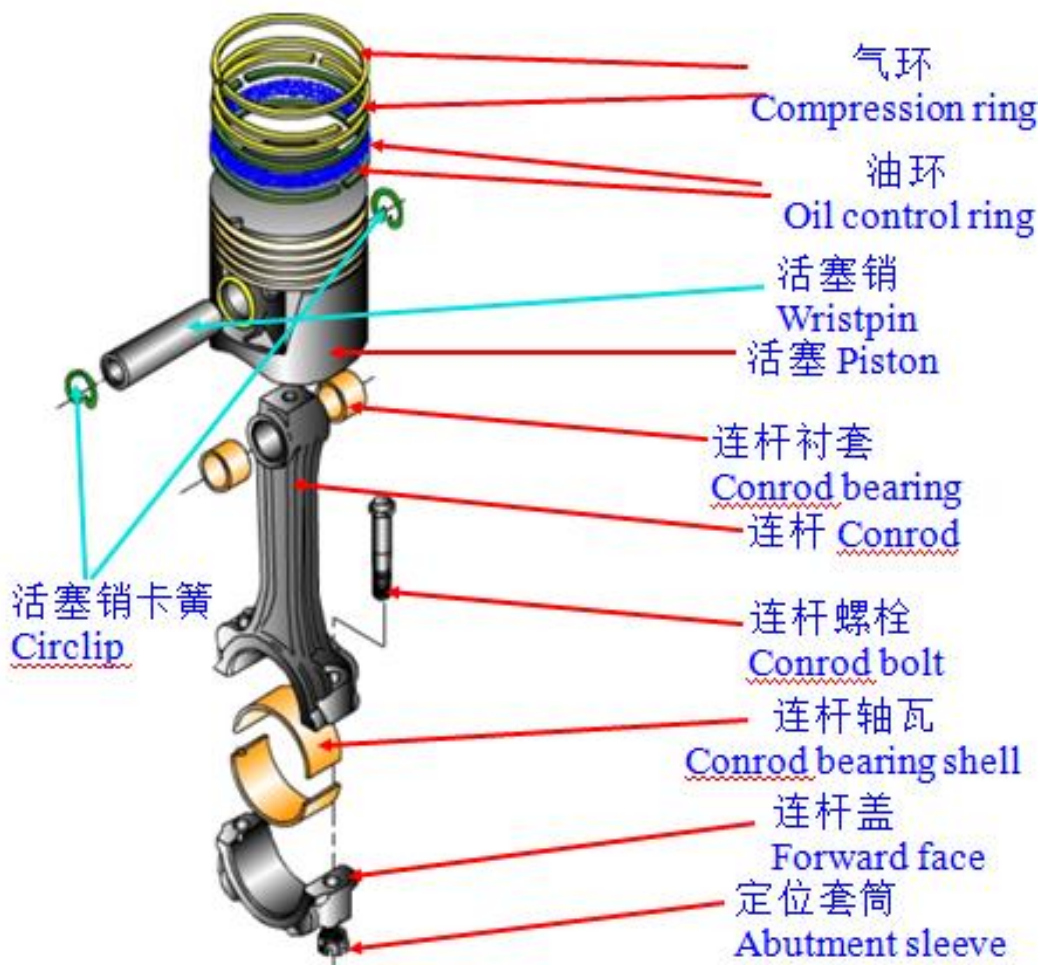
油底壳

The oil supply for passenger car engines today is provided almost exclusively with a wet sump lubrication design. In such engines the oil pan forms the bottom termination for the engine block.

Function:

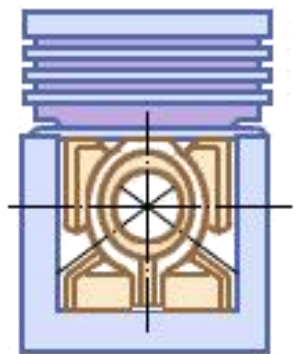
- Serve as a container to receive motor oil and as the collecting basin for oil returning from bearings and lubrication points
- Enclosing the crankcase and serving to stiffen the engine and transmission assembly
- Taking threads for oil drain plug and dipstick guide tube, usually equipped with magnet to absorb metal filing

2.5 Piston-Rod Assemblies (活塞连杆组)

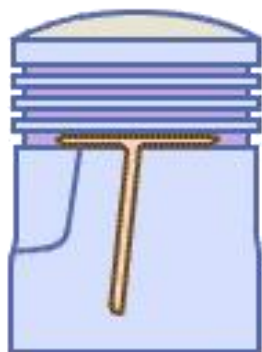


- Function:
 - ✓ Accepting the pressures created by combustion
 - ✓ Transferring forces via the wristpin and the connecting rod to the crankshaft
 - ✓ Providing guidance for the small conrod eye (连杆小头)
- Requirements for the piston:
 - ✓ High static and dynamic strength at high temperatures, little plastic deformation (弹性形变)
 - ✓ Low weight, resulting in small inertial forces
 - ✓ Low friction resistance, high wear resistance (耐磨性能)
 - ✓ Low noise, no “piston slapping” with engine cold and warm

2.5.1 Piston Crown and Piston Head (活塞顶及活塞头)



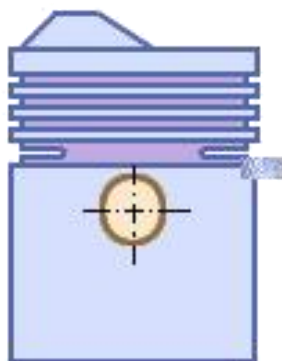
平顶活塞



凸顶活塞



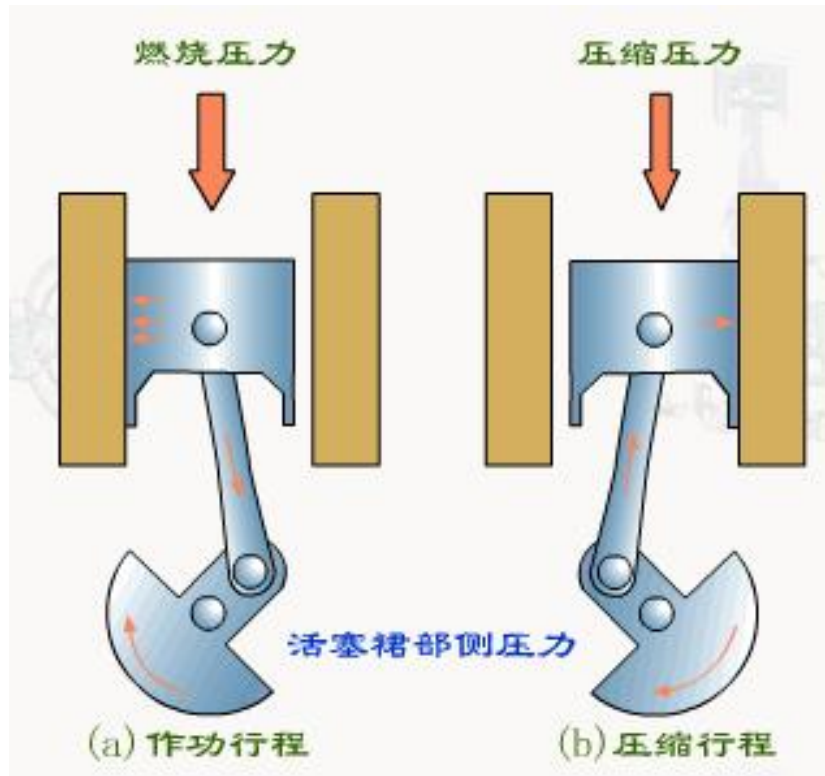
凹顶活塞



成型顶活塞

- Piston crown is the lower half of combustion chamber, and is directly exposed to gas forces. The shape and size of piston crown strongly affect the mixture formation (混合气形成) and combustion process
- The shape of piston crown could be divided into four types: flat coping piston, convex coping piston, concave coping piston and model piston
- Piston head refers to the part from the first piston groove (活塞环槽) to the wristpin. Three or four grooves are machined into each piston for positioning the compression rings (气环) and oil rings (油环)

2.5.2 Piston Skirt (活塞裙)



- Piston Skirt refers to the part from oil ring groove to the bottom of the piston
- Function:
 - ✓ Used to guide the piston within the cylinder.
 - ✓ Transfers to the cylinder wall, in sliding fashion, the lateral forces occurring because of the deflection (偏移) of the conrod.
 - ✓ With sufficient skirt length and close guidance the so-called “piston slapping”, occurring at the moment when contact shifts from the one side of the piston to the opposite side, is kept to a minimum. This is important for smooth engine running and to reduce wear at all the piston’s sliding surfaces.

2.5.3 Piston Skirt Structure

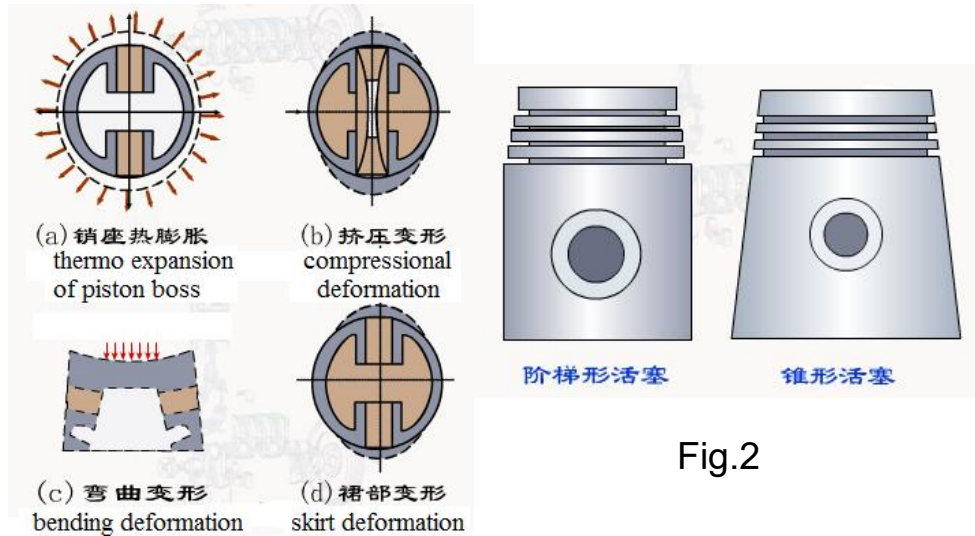


Fig.1

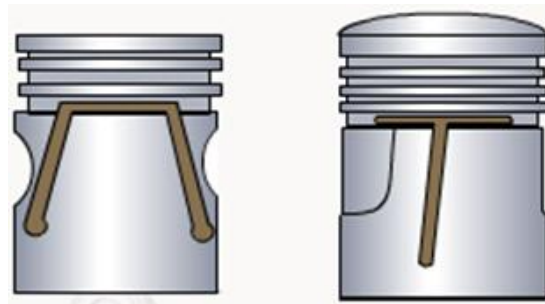


Fig.3

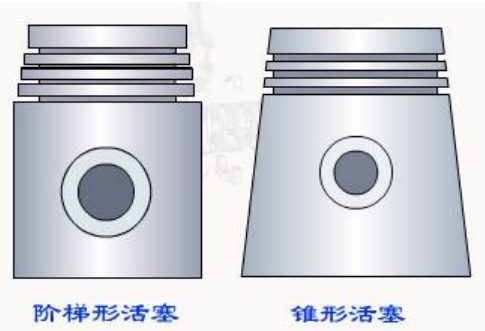


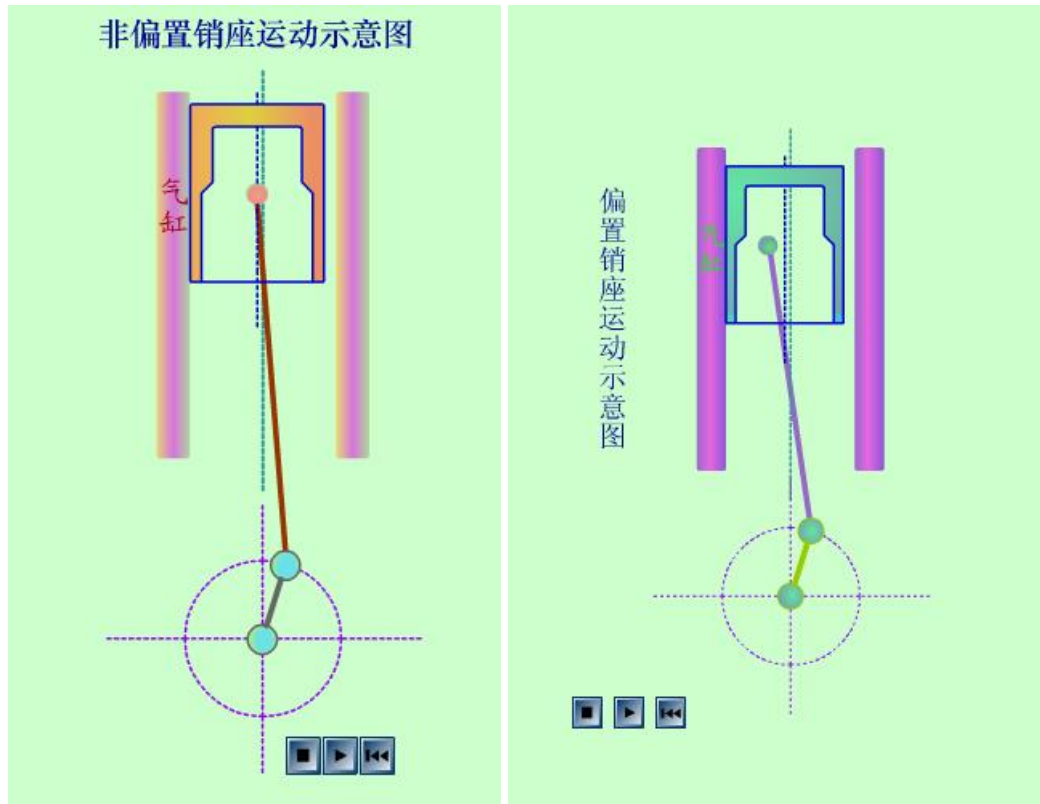
Fig.2



Fig.4

- In order to compensate with varies type of skirt deformation, stricture of piston skirt is complicated:
 - ✓ Oval-shaped (椭圆形) (radial)
 - ✓ Ladder or corn shaped (axial)(Fig.2)
 - ✓ Slotting near the piston skirt (Fig.3)
 - ✓ Cutting off part of skirt which is not subjected to force (less weight, suitable for higher engine speed) (Fig.4)
 - ✓ Bimetetallic Piston (双金属活塞)

2.5.3 Piston Skirt Structure



a) 活塞销对中布置

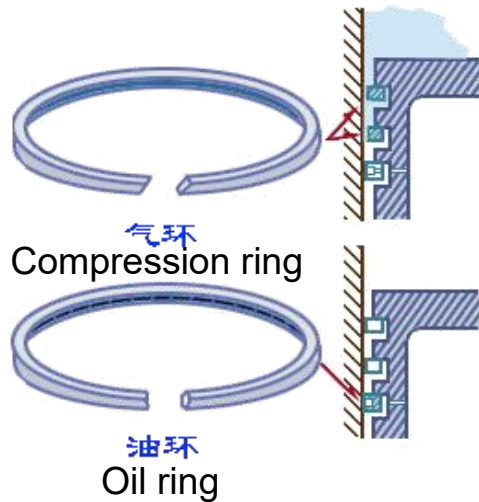
Piston pin without deaxising

b) 活塞销偏移布置

Piston pin deaxising

- Axially offsetting (轴向偏心) in the direction of pressure (direction in which the piston contacts the cylinder barrel (气缸筒) in the expansion stroke) causes an earlier contact change for the piston when the normal force on the piston is weaker.
- The tilting (倾斜) movement of the piston causes it to first contact the cylinder with the “soft” bottom part (piston skirt), which reduces impact, therefore reduce noise.

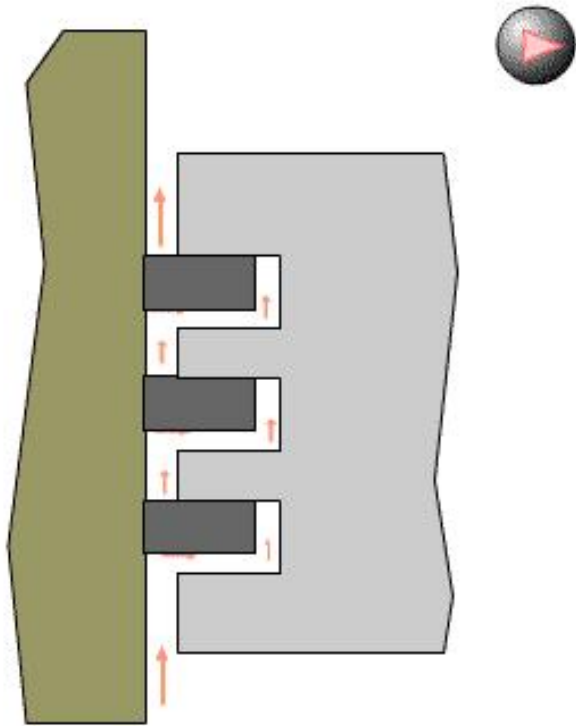
2.5.4 Piston Ring (活塞环)



- Piston rings are metallic gaskets whose functions are to seal the combustion chamber against the crankcase (曲轴箱), to transmit heat from the piston to the cylinder wall (70%~80%), and to regulate the amount of oil present on the cylinder sleeve, a function of the oil control ring in particular.

- Loading, Damage, Wear, Friction and materials for Piston Ring
 - ✓ Piston rings are loaded by outward stresses when installed, the outward stress is equal to the strain imparted when the ring is passed over the cylinder.
 - ✓ Dynamic loading occurs in addition an axial motion of the piston ring caused by the interactions of gas, mass, and friction forces, which leads to ring chatter and vibration.
 - ✓ Extraordinarily high loading on the ring can arise from soot (积碳) collecting
 - ✓ Materials: cast iron(铸铁) with flaked graphite (片状石墨), cast iron with spheroidal graphite (球状石墨), steel

2.5.4 Piston Ring (活塞环)



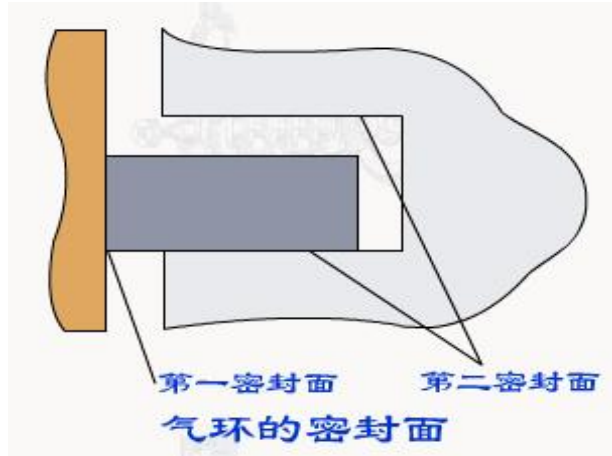
气环的泵油作用

- **Pumping effect :**

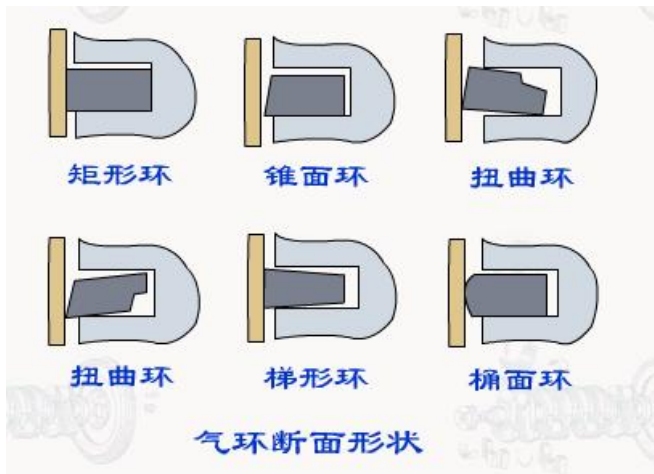
Rectangular Compression ring will deliver lubrication oil into the cylinder during oscillating movement

- ✓ Increase consumption of lubrication oil
- ✓ Wet the spark plug
- ✓ Increase the deposit inside the cylinder and the piston ring groove
- ✓ Intensify wear and tear of the compression ring

2.5.4 Piston Ring (活塞环)



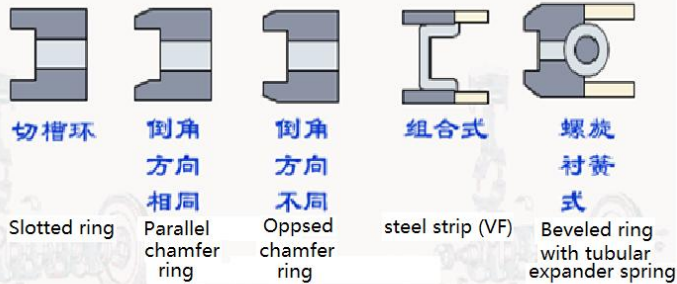
- Compression Rings
 - ✓ Close contact with both cylinder wall and the flank (侧面) of the groove machined into the piston
 - ✓ Cylinder wall contact—spring action inherent to the ring itself
 - ✓ Ring groove contact— gas pressure in the combustion chamber
 - ✓ Two rings for SI engines, three rings for CI engines for higher cylinder pressure
 - ✓ Types of compression rings: Rectangular ring (矩形环), Bevel ring (锥面环), ring with inside shoulder (twist beveled ring) (扭曲环), double trapezoid ring (梯形环), spheroid casting ring (桶面环)



2.5.4 Piston Ring (活塞环)



- Oil Control Rings
- Oil control rings are of particular significance in managing the engine's oil supply and consumption; they are subdivided into the following:
- Self-expanded cast iron rings (control action in high-speed engines and as the only oil control ring on a piston)
- Spring-expanded and spring-backed oil control rings (manufactured as castings, for gasoline and diesel engines)



different types of oil control rings

2.5.5 Wristpins(活塞销)

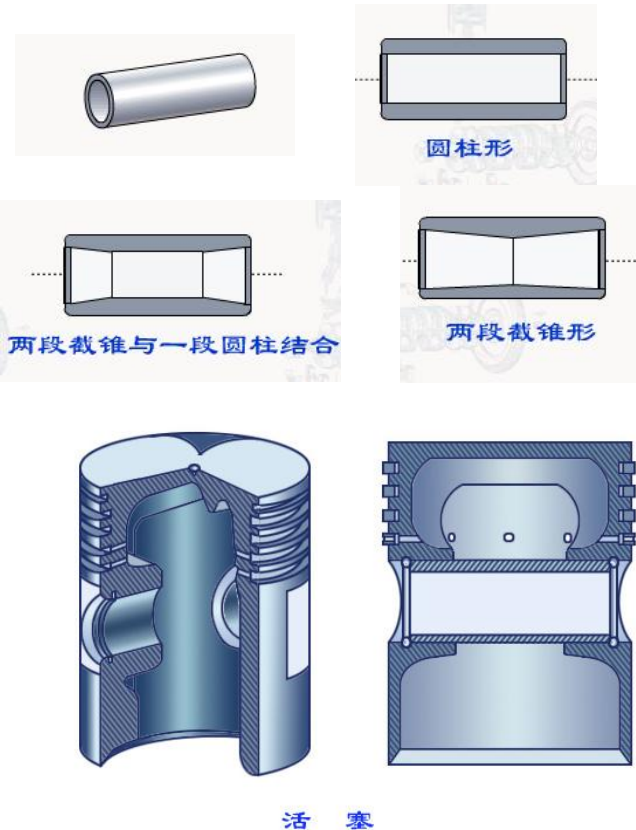
- Functions:

The wristpin makes the connection between the piston and the connecting rod. It is subjected to the extreme, alternating loads exerted by the pressure of the exploding gas and the mass inertias. The lubrication situation is unfavorable because of the rotary motions between the piston and the wristpin and between the wrist pin and the conrod.

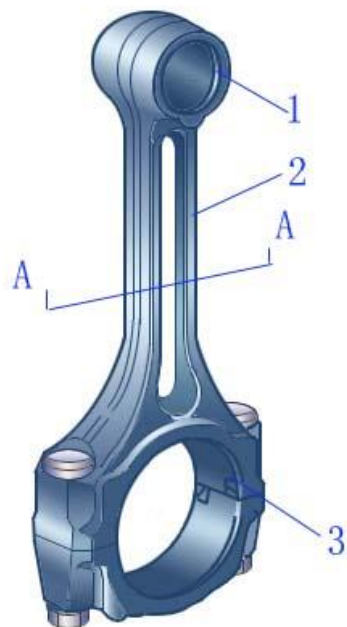
- Designs:

✓ The wristpin with cylindrical inside and outside contours has been successful in most applications. To reduce weight and the mass inertias, the out ends of the wristpins' inside bore may be conical (圆锥形) since the load is less there.

✓ Wristpins in passenger car gasoline engines are often held in the conrod with a tension due to shrinkage (收缩) (shrink wristpin). In more heavily loaded gasoline and diesel engines, the wristpin “floats” in the conrod. It is secured with circlips (弹簧挡圈) to keep it from wandering laterally and out of the piston



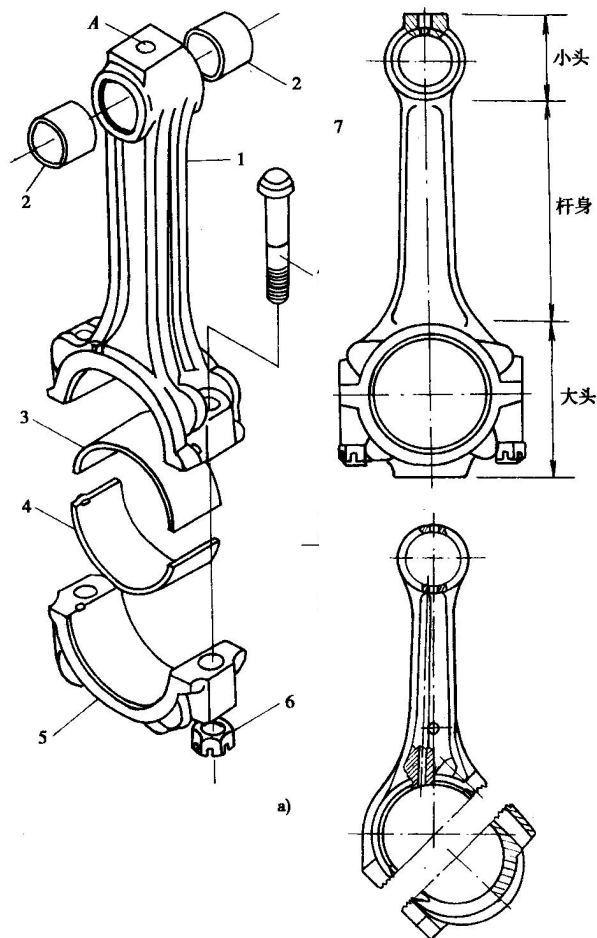
2.5.6 Connecting Rod (Conrod)(连杆)



A-A 剖面

- The connecting rod has two so-called conrod ends.
- It is at the small conrod eye (连杆小头) that the connection to the piston is made by the wristpin.
- The split, large connecting rod eye (连杆大头) is located at the crankshaft end of the rod. Proper functioning is ensured with a sliding bearing and by fixing and screwing down the conrod bearing cap (轴承盖).
- The connecting rod shaft joins the two connecting rod eyes. This section may have a special cross section depending on the requirements ,e.g., *I shaped* or *H shaped*.
- Depending on the particulars of the application and the resultant loads, a number of different materials may be used for connecting rods: Cast iron (铸铁) , Forging steel (锻钢), Powdered metal (金属粉末成型) and alternate materials such as carbon fiber reinforced aluminum or carbon fiber reinforced plastic.

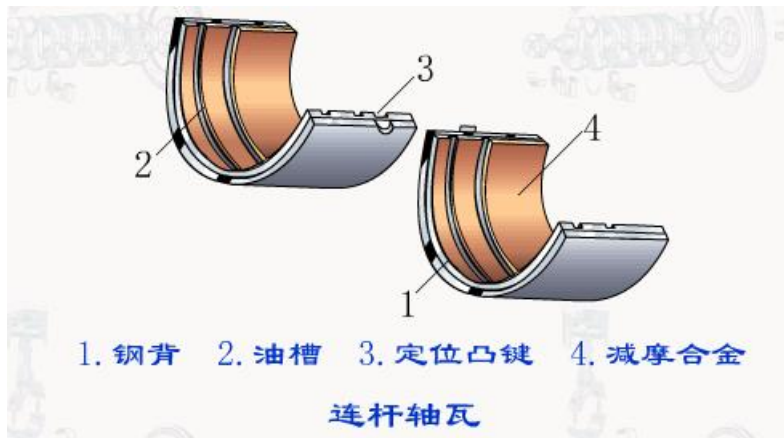
2.5.6 Connecting Rod (Conrod)



- Conrod structure:
 - 1 Shaft (连杆体)
 - 2 Conrod bearing (连杆衬套)
 - 3-4 Conrod bearing shell(连杆轴瓦)
 - 5 Conrod cap(连杆盖)
 - 6-7 Bolt (连杆螺栓)
- A Oil bore
- Big-end eye is normally split perpendicular to the long axis of the conrod. It could also be split at an angle to reduce the maximum width of the conrod
- Conrod fracture splitting (连杆大头胀断) describes the separation of the conrod web and the cap by breaking the latter away during processing:
 - ✓ Reducing the number of processing steps
 - ✓ Secured against relative movement with the irregular surfaces
 - ✓ Eliminating the need for any additional guide elements
 - ✓ Easy matching for multi-cylinder engines

2.5.7 Conrod bearing shell(连杆轴瓦)

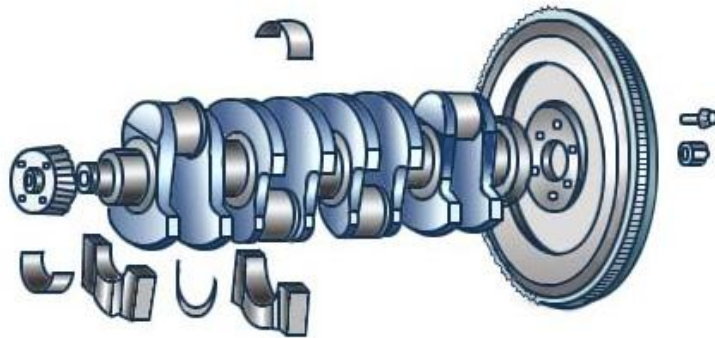
- The shafts found in multi-cylinder reciprocating engines generally run in plain bearings. The reasons for selecting this type include their great ability to withstand shock and their damping properties, easy division for assembly around the crankshaft, low space requirements, insensitivity to grime (污垢), and, last but not least, the low costs when compared with rolling bearings (滚动轴承). However, it has higher friction level and the resulting greater oil requirements.



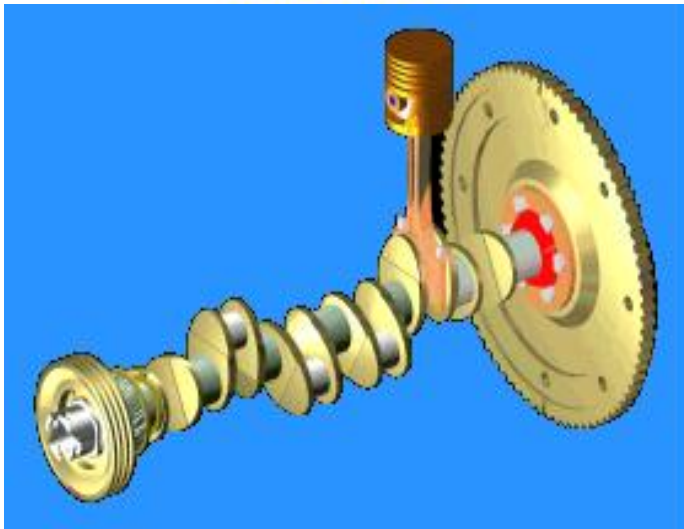
- **Material:**

- ✓ **Babbitt metals(巴氏合金)**
- ✓ **Aluminum Alloys**
- ✓ **Copper Alloys**

2.6 Crank Shaft-Fly Wheel Assemblies(曲轴飞轮组)



曲轴飞轮组



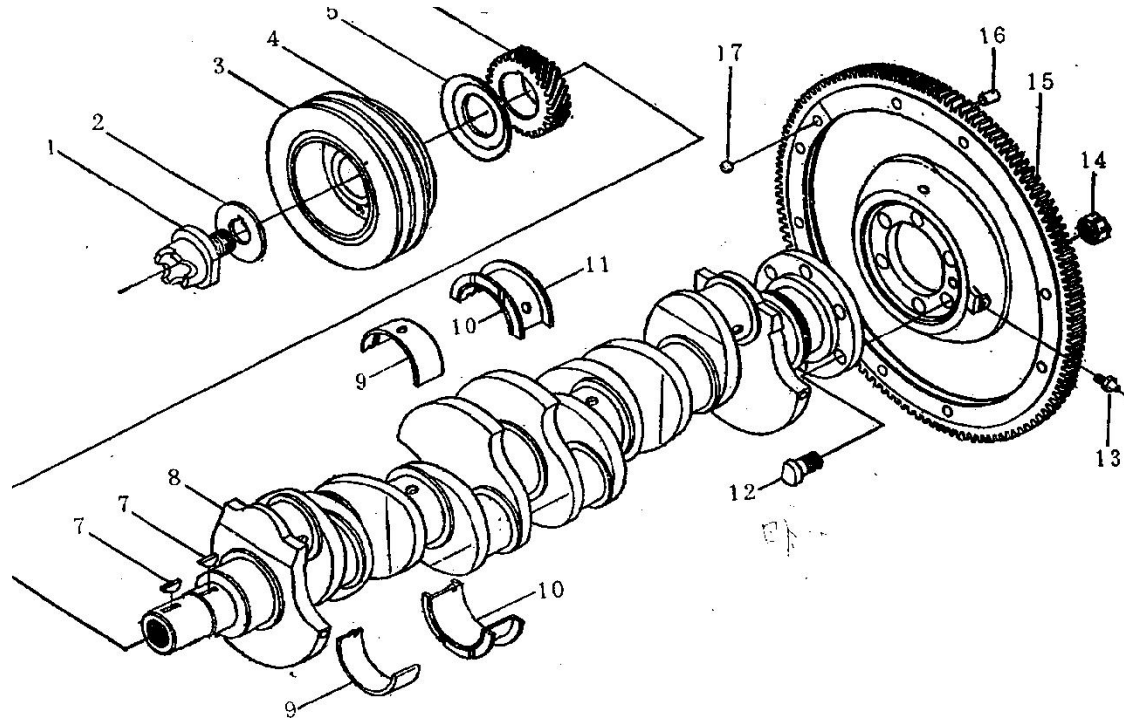
Crankshaft Drive (曲轴)

•Function:

Crankshaft drive is a functional group that not only efficiently transforms oscillating movement(back-forth movement) into rotary movement, but is also excellent at converting thermodynamic processes to yield the maximum work, efficiency, and technical feasibility. These advantages are gained at cost of serious disadvantages:

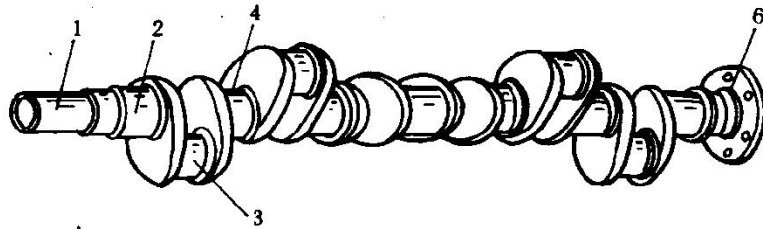
- ✓ Limitation of speed due to free inertia
- ✓ Uneven force transmission
- ✓ Great deal of stress on the crankshaft and high fluctuations
- ✓ Tribological (摩擦学) problems

2.6 Shaft-Fly Wheel Assemblies(曲轴飞轮组)

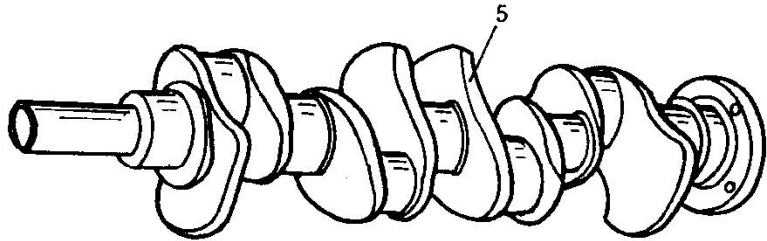


- The crankshaft drive in automotive engines consists of pistons with rings, piston pins, conrods, a crankshaft with countermass, crank shaft main bearing, and the lubricant

2.6.1 Crankshaft(曲轴)



a) 解放CA6102型发动机曲轴



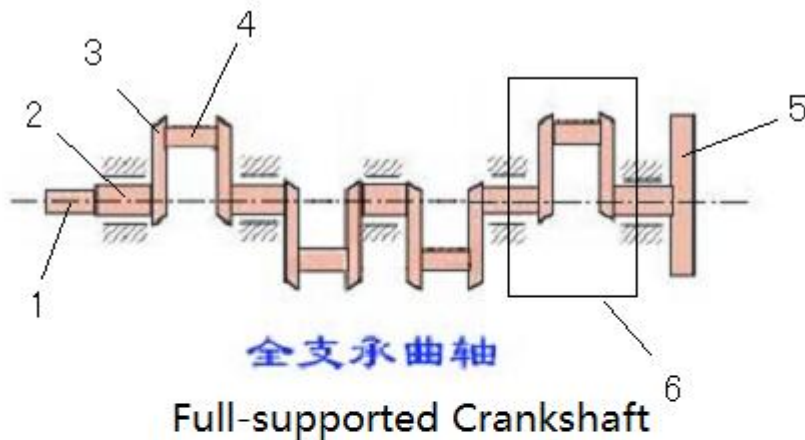
b) 北京BJ492型发动机曲轴

图 2-48 曲轴

1-前端轴；2-主轴颈；3-连杆轴颈（曲柄销）；
4-曲柄；5-平衡重；6-后端凸缘

- Function: The piston's linear movements are converted, with intervention of the conrod, into rotational movement at the crank, thus making torque available for use at the wheels
- Loadings: Because of the stains involving forces that change in both time and location, with rotational and flexural (弯曲) torques and the resulting excitation for vibrations, the crankshaft is subject to very high, very complex loads
- Components: Crankshaft front end(1), Crankshaft main journal(2), crank pin(3), crank(4), counterweight(5), back-end crank(6)
- Number of crank pins determines the number of cylinders

2.6.2 Crankshaft support pattern(曲轴支承方式)



- Crankshaft could be divided into types by different support pattern, the full-supported crankshaft and partly-supported crankshaft
- The Full-supported type has a better stiffness and bending strength, therefore is used widely in heavy duty engines. However, finished surfaces and number of main bearings of this type of Crankshaft is more than that of the partly-supported type, and the length of engine would also be longer



2.6.3 Mass Balancing (平衡重)

- Mass balancing is the compensation of imbalances due to construction. The rotating inertial force (惯性力) can be balanced by counterweights (配重) where the static torque of the rotating masses and the balancing masses must correspond.

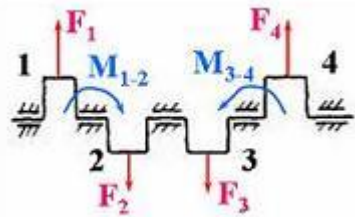


Fig.1 Force balance

(受力平衡)

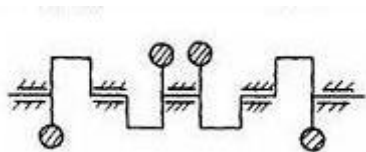


Fig. 2 Mass balance

(设置平衡重)

- It is shown in Fig.1 that the force on the crankshaft has balanced with each other, but the torque couple M_{1-2} and M_{3-4} generate bending load to the crankshaft. Therefore, counterweights are employed to improve the loading condition of the crankshaft
- Counterweights would increase the total mass of crankshaft, therefore increase the centrifugal force and rotating inertial force, which would put more stress on bearing shells

2.6.4 Throw and Firing Sequences(曲柄及气缸发火顺序)

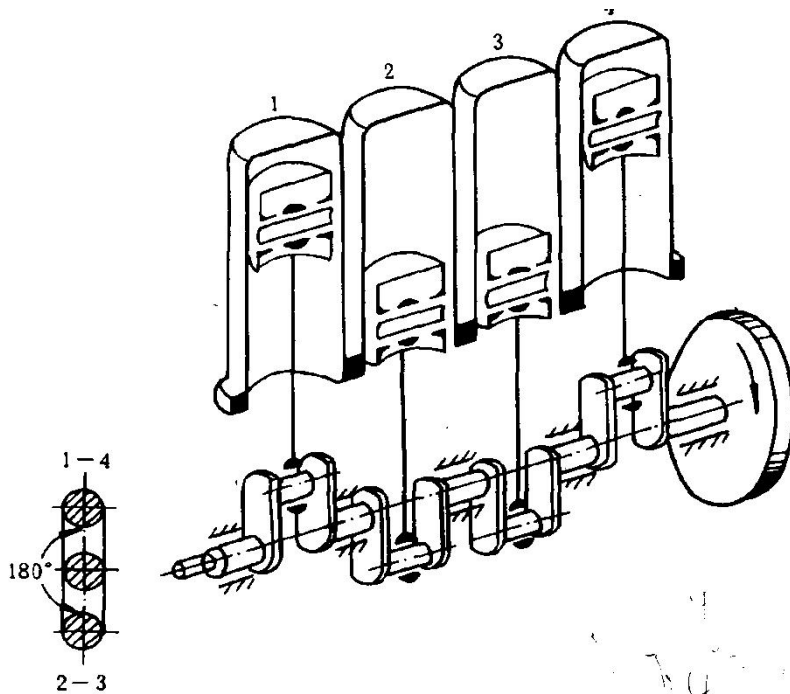


图 2-55 直列四缸发动机的曲拐布置

- To obtain a very even torque characteristic, the ignition of the individual cylinders must be evenly distributed over the power cycle. A requirement is that the throws (曲拐) must be evenly distributed over the perimeter (周长). Hence, the following throw spacing results:

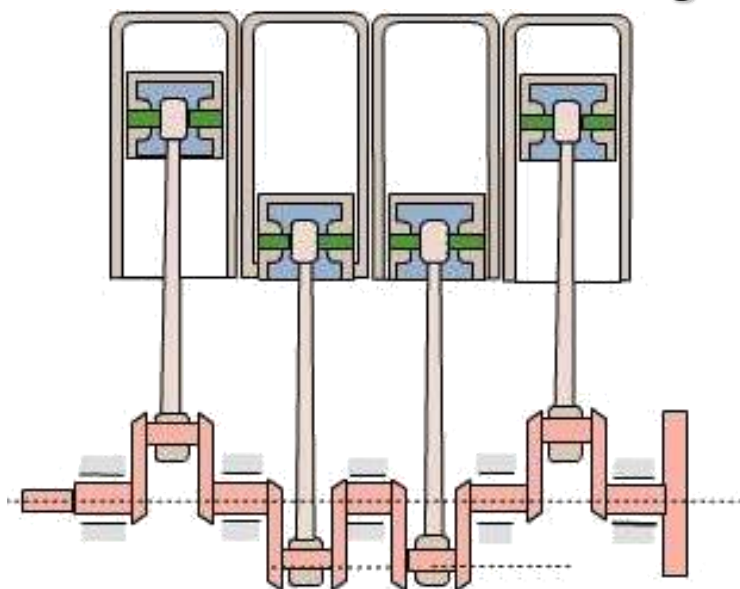
Four-stroke engines: 720° crankshaft/cylinder number

- The firing sequence is also determined by the direction of rotation of the crankshaft. Other factors for the firing sequence are:

- ✓ No or very small free inertial effects
- ✓ Favorable rotational oscillation behavior
- ✓ Good supercharging conditions

- Take a inline four-cylinder engine for example, favorable firing sequence could be 1-2-4-3 or 1-3-4-2

2.6.4 Throw and Firing Sequences(曲柄及气缸发火顺序)



四缸机工作循环表

(发火次序:1—2—4—3) 表 2-1

曲轴转角(°)	第一缸	第二缸	第三缸	第四缸
0~180	作功	压 缩	排 气	进 气
180~360	排 气	作功	进 气	压 缩
360~540	进 气	排 气	压 缩	作功
540~720	压 缩	进 气	作功	排 气

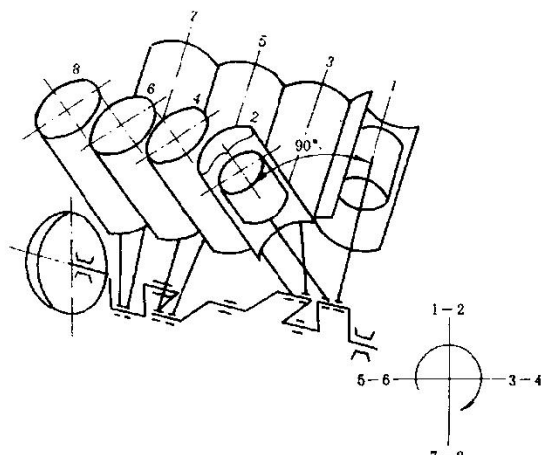


四缸机工作循环表

(发火次序:1—3—4—2) 表 2-2

曲轴转角(°)	第一缸	第二缸	第三缸	第四缸
0~180	作功	排 气	压 缩	进 气
180~360	排 气	进 气	作功	压 缩
360~540	进 气	压 缩	排 气	作功
540~720	压 缩	作功	进 气	排 气

2.6.4 Throw and Firing Sequences(曲柄及气缸发火顺序)



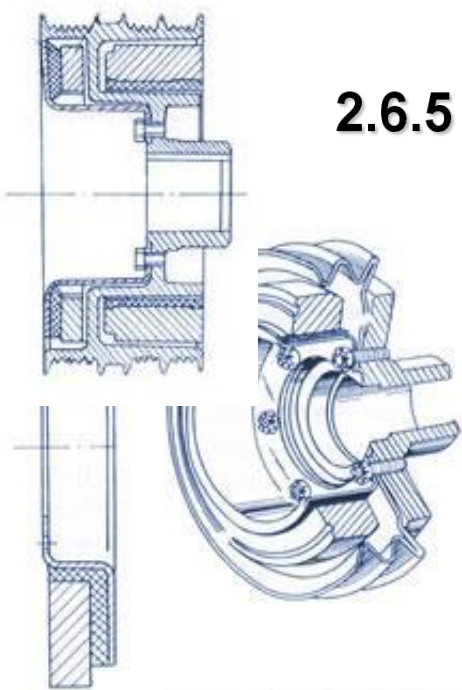
- Another example for a V8-engine, throw spacing is $720^\circ / 8 = 90^\circ$
- Two cylinders at the same role share the same crank
- Firing sequence is 1-8-4-3-6-5-7-2

V 型八缸机工作循环表

(发火次序: 1—8—4—3—6—5—7—2)

表 2-4

曲 轴 转 角 ($^\circ$)	第一缸	第二缸	第三缸	第四缸	第五缸	第六缸	第七缸	第八缸
0	作 功	作 功	进 气	压 缩	排 气	进 气	排 气	压 缩
0~180								
90	排 气	排 气	压 缩	作 功	进 气	压 缩	进 气	作 功
180								
180~360	进 气	进 气	作 功	排 气	压 缩	作 功	压 缩	排 气
270								
360	压 缩	压 缩	排 气	进 气	作 功	排 气	作 功	压 缩
360~540								
450	作 功	作 功	进 气	压 缩	排 气	进 气	排 气	压 缩
540								
540~720	排 气	排 气	压 缩	作 功	进 气	作 功	排 气	作 功
630								
720								



2.6.5 Torsion vibration damper(扭转减震器)

- Without damping, the excursions of the crank shaft would become increasingly larger until the shaft breaks, therefore, additional measures must be taken, and the only possibility is to reduce the rotational oscillations to a safe level, there are basically two options for this: Damping(减幅) and Absorption(吸收).
- The effect of vibration dampers are based on both damping and absorbing. They are designed to continuously reduce rotational oscillation excursions of the system.
- For passenger car engines, rubber vibration dampers are used: the vibration energy is converted by the material damping of the rubber into heat. Two stage dampers are also used in which two damper masses are tuned to two different frequencies.
- The dampers not only relieve the crankshaft and camshaft mechanically, but also reduce the play-induced noise of the engine.

camshaft rubber vibration dampers

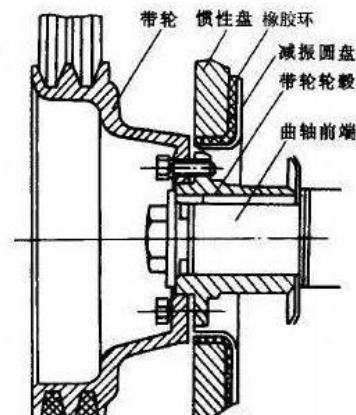
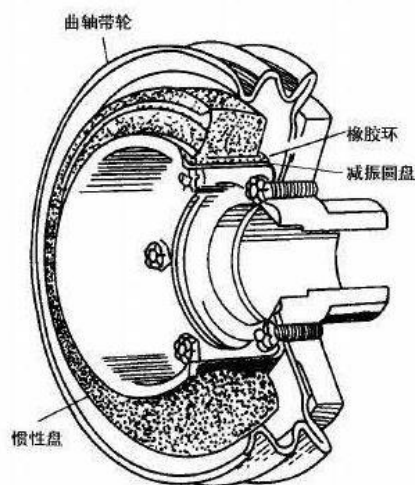
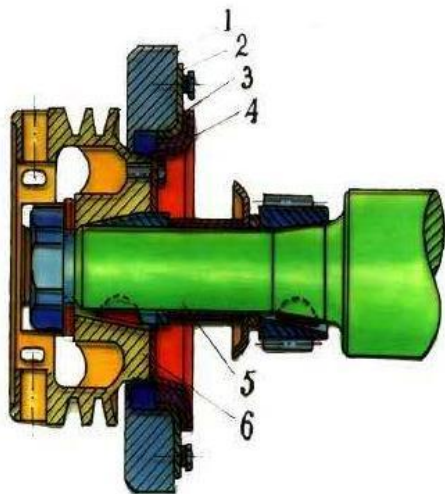
功用：降低曲轴扭转振幅,防止曲轴共振

将减振圆盘用螺栓与曲轴皮带轮紧固在一起，橡胶层与圆盘及惯性盘硫化在一起。当曲轴发生扭转时振动时，力图保持等速转动的惯性盘便使橡胶层发生内部摩擦，从而消除了扭转振动的能量，避免扭振。

橡胶摩擦式扭转减振器

结构：惯性盘、减振圆盘、橡胶层

原理：惯性盘与减振圆盘相对位移，橡胶层扭转吸收能量



Torsion vibration damper (cases)

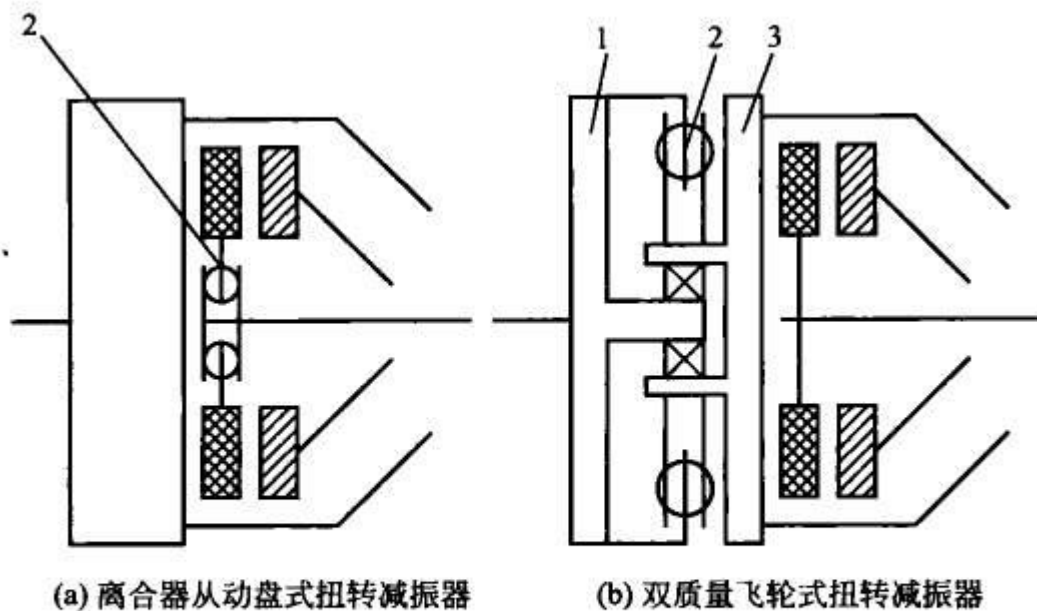


图 2.21 两种扭转减振器的结构比较示意图

1—第一飞轮；2—减振器；3—第二飞轮

2.6.6 Flywheels(飞轮)



- A flywheel is a rotating mechanical device that is used to store rotational energy. It has a significant moment of inertia and thus resists changes in rotational speed. It is often used to provide continuous energy in systems where the energy source is not continuous.
- It is used to maintain constant angular velocity of the crankshaft in a reciprocating engine. In this case, the flywheel stores energy when torque is exerted on it by a firing piston, and it releases energy to its mechanical loads when no piston is exerting torque on it, therefore enhance the distribution of the rotation speed and torque output.
- Drives the actuator(驱动件) of the clutch (离合器)
- Flywheels have ring gear at the outer rim of the wheels and is geared with starting motor to start the engine

Homework to review

1. Function of the components for the cylinder head, block and their configurations?
2. What are the main parts of Crank and Connecting Rod Mechanism?
3. How many types of Cylinder Block configurations?
4. What is the piston slapping? How to avoid this in application?
5. What is the oil pumping of piston rings? How to solve it?
6. What is the function of Flywheel?
7. How to solve the torsion vibration of crankshaft?